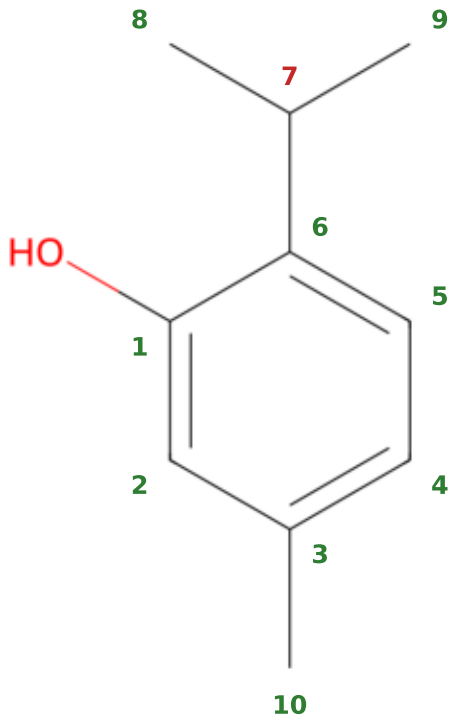
Thymol: predicted vs experimental ^{13}C shifts

SMILES (from Thymol.svg): `Cc1ccc(C(C)C)c(O)c1`

Prediction: CASCADE-2.0 Exp22K-FF-GPR, Boltzmann-weighted MMFF ensemble (1 conformer); symmetry-equivalent carbons averaged

Experimental: 13C NMR (126 MHz): δ 20.9, 22.7, 26.7, 116.0, 121.7, 126.3, 131.3, 136.6, 152.5.

Counts agree: 9 predicted environments = 9 experimental peaks



High confidence Moderate confidence Low confidence

Labels: carbon number, coloured by prediction confidence

Carbon(s)	Exp δ (ppm)	Calc δ (ppm)	Δ calc−exp	σ (ppm)	Confidence
10	20.9	20.88	-0.02	0.720	High
8,9	22.7	22.76	+0.06	0.718	High
7	26.7	26.73	+0.03	0.775	Low
2	116.0	116.44	+0.44	0.700	High
4	121.7	121.79	+0.09	0.690	High
5	126.3	126.42	+0.12	0.698	High
6	**131.3**	**131.76**	**+0.46**	**0.716**	High
3	136.6	136.47	-0.13	0.694	High
1	152.5	152.66	+0.16	0.710	High

MAE = 0.17 ppm over 9 matched pairs; max |Δ| = 0.46 ppm (highlighted row)

Rows are ordered by ascending experimental shift. Predicted environments were assigned to experimental peaks one-to-one so that the total |Δ| (and hence the MAE) is minimal; this is a best-fit assignment, not a verified spectral assignment. σ is the standard deviation of the model's Gaussian-process prediction; confidence uses the CASCADE web-server thresholds on 1.96σ (High < 1.42 ppm ≤ Moderate < 1.49 ppm ≤ Low). Carbons are numbered 1, 2, 3 ... in their order in Thymol.svg (carbons only).